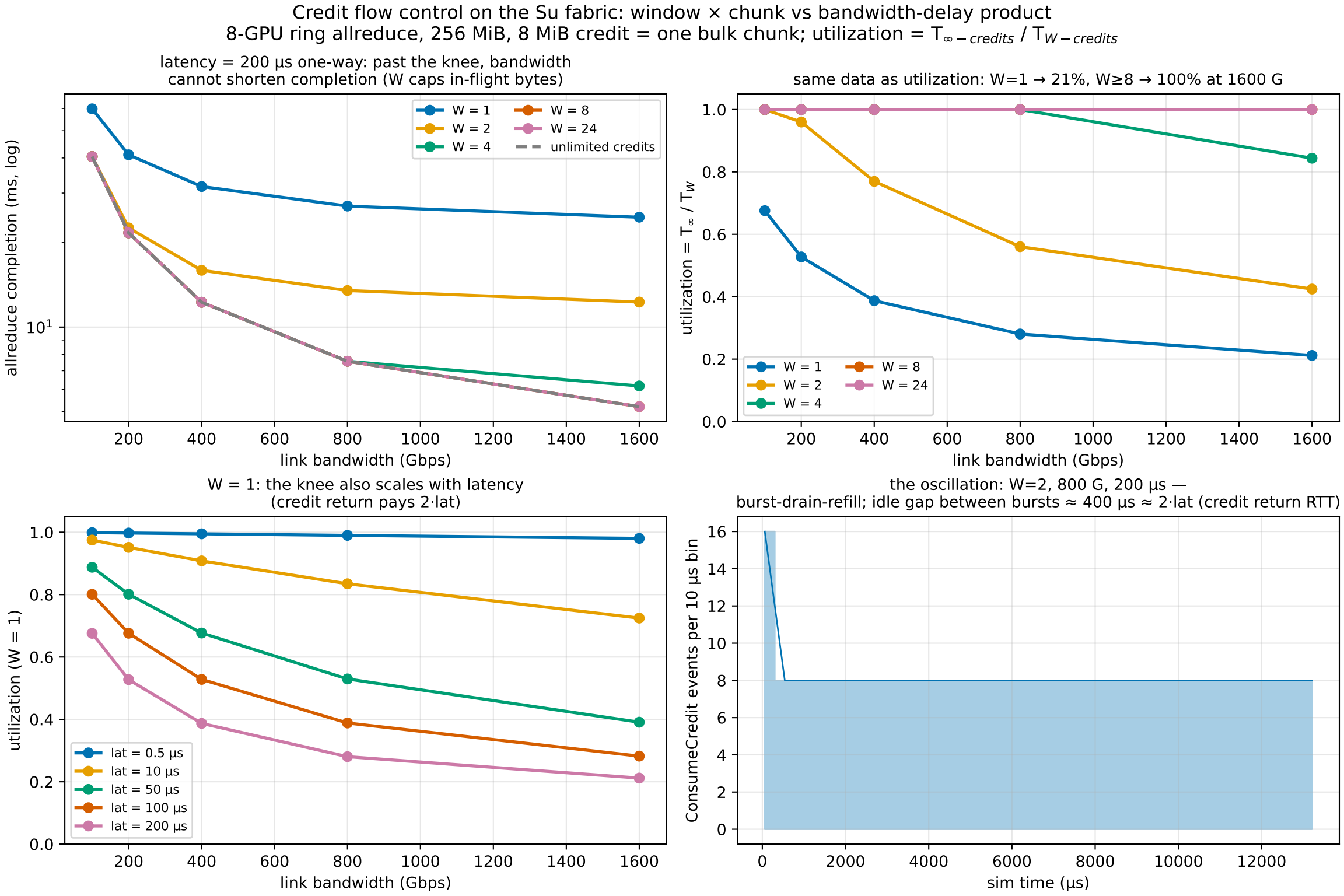
Credit flow control on the Su fabric: window × chunk vs bandwidth-delay product
8-GPU ring allreduce, 256 MiB, 8 MiB credit = one bulk chunk; utilization = T∞ − credits / TW − credits
latency = 200 µs one-way: past the knee, bandwidth
cannot shorten completion (W caps in-flight bytes)
allreduce completion (ms, log)
link bandwidth (Gbps)
W = 1
W = 2
W = 4
W = 8
W = 24
unlimited credits
same data as utilization: W=1 → 21%, W≥8 → 100% at 1600 G
utilization = T∞ / TW
W = 1: the knee also scales with latency
(credit return pays 2·lat)
utilization (W = 1)
lat = 0.5 µs
lat = 10 µs
lat = 50 µs
lat = 100 µs
lat = 200 µs
the oscillation: W=2, 800 G, 200 µs —
burst-drain-refill; idle gap between bursts ≈ 400 µs ≈ 2·lat (credit return RTT)
ConsumeCredit events per 10 µs bin
sim time (µs)